Lab 8: Kick start group project

Overview

In this lab, we will kickstart the group projects.

To receive credit for this lab, you MUST show your work to the TA during the lab, and push it to the github before the deadline. Please note that submissions will be due on Wednesday at 11:59 p.m. in the following week.

> **UPDATED SUBMISSION GUIDELINES**
>
> From this lab onwards, at least ONE PERSON PER TEAM must submit a link to the lab's deliverable to Canvas. This **guide** should serve as a guide to create a repository on GitHub.

Learning Objectives

All of these tie in closely with the group project.

☐ LO1. Learn how to structure an application with a GUI, database, and a middle layer connecting the UI to data, while optionally integrating APIs.
☐ L02. Gain practical experience in containerizing applications using Docker to ensure portability and consistency across different environments.
☐ L03. Develop skills in applying Agile or hybrid methodologies, such as Scrum and Kanban, to manage project tasks and team collaboration effectively.
☐ L04. Learn to design high-level use case diagrams and wireframes to define the application's key features, user interactions, and interface layout.

> **PROJECT GUIDE**

> Please read the **project guide** alongside this writeup to get more information on the expectations for this group project

Part A

In today's lab, you will work on starting off your project. Your TA will inform you of the approved project idea. You will be working with your team of 4-6 students on this project for the next 4 weeks. You will submit the project proposal by the end of this lab.

> **MANAGING PROJECT REPOSITORY**
>
> Only one person per team should create the project repository. This **guide** has instructions on adding collaborators to a repository. Your project repository has to be **public.** If you wish to keep it private, please reach out to your TA or instructor.

Project Overview

Your application must have a GUI, a database, API data source(optional), a middle layer connecting your UI to your data. The code should be built within Docker containers so they are portable. Make sure to consider these core requirements when designing the architecture of your application. You are **required** to use the technology stack that you have learned in this course. Refer to the **project guide** for detailed expectations. If you would like clarifications, please ask your instructor/TA.

Deliverables

Before starting this discussion, assign a scribe who will start a document and take notes on the items listed below. The team will proofread and submit this document as your deliverable for this lab.

1. Team Number

You have been assigned a team number, on Canvas, that is represented as your recitation section number plus a one-digit identifier. This team number MUST appear on ALL deliverables/submissions posted by your team to your GitHub repository.

2. Team name

You will collectively choose a name for your team to represent a vision you have for the application you're creating. It can be used in addition to your Team Number for all deliverables and presentations.

3. Team Members

List the First & Last name of team members, their GitHub usernames and their email addresses.

4. Application Name

Pick a befitting name to convey the purpose of your application. To be used in all presentations(weekly to your TA and the final one)

5. Application Description

A short (2-3 paragraphs) overview description of the application. Provide enough information to explain what functionality and value your product will provide to users of your application. This should help you to define the scope of your project.

6. Audience

Define the target audience for your application by identifying the characteristics of an ideal user. Clearly articulate the specific problem your software aims to solve for this user group. Additionally, consider the usability and accessibility of the application, ensuring that interactions are intuitive and convenient for the intended audience.

Understanding their background helps shape features and design.

7. Vision Statement

A simple, one or two sentence statement describing the clear and inspirational desired state resulting from your team’s efforts to create your application. Following is a template you might want to use:

For [target customer], **Who** [statement of the need]. **The** [product name] **is a** [product category] **that** [key benefit, compelling reason to buy]. **Unlike** [primary competitive alternative], **our product** [statement of primary differentiation]

Examples

- **Amazon**: "To be Earth's most customer-centric company, where customers can find and discover anything they might want to buy online."
- **Tesla**: "To create the most compelling car company of the 21st century by driving the world's transition to electric vehicles."
- **Facebook**: "People use Facebook to stay connected with friends and family, to discover what's going on in the world, and to share and express what matters to them."
- **CU Boulder**: "To be a leader in addressing the humanitarian, social, and technological challenges of the twenty-first century."
- **IKEA**: "To create a better everyday life for many people."
- **Nordstrom**: "Offer the customer the best possible service, selection,quality, and value."
- **Google**: "To provide access to the world's information in one click."

8. Version Control

> **INFO**
>
> You are required to create a **public** GitHub repository and add all the members of the team to it. Share the link to this repository in this document.

The root of your project repository should ONLY have the following files/folders. Make sure you organize any additional files you create for the project within the folders listed below.

- **TeamMeetingLogs** - The minutes of the meeting with your TA every week will be recorded in a file here. You will be updating the same file every week with the latest updates. Points to include in the minutes would be:
  i. Decisions made
  ii. Alternative actions/options discussed
  iii. Follow-up items, including agreed-on roles and responsibilities
- **MilestoneSubmissions** - All course-related documents, including this one, will be stored in this folder.

- **ProjectSourceCode** - The source code and all relevant project documentation for the application will be stored in this folder.
- **ReadMe.md** - Refer to the **project guide** for more information on this.
  - For a detailed view of your repository, refer to the **project guide**
- **.gitignore** - Please remember to create a .gitignore file in your "ProjectSourceCode" folder in the repository

9. Development Methodology

Since this project is scheduled for 4 weeks, we encourage you to use an Agile Methodology. You can also use a hybrid set of practices from different methodologies. For example, you can use Scrum meetings and Kanban boards to track the progress of your project. The important thing is to have a plan and stick to it.

10. Communication Plan

Describe in a sentence or two how your team plans to communicate with each other during the course of the project. You must identify a collaboration tool for team members to utilize for coordination of their work and communication among team members and with the TA. Some tools you could use: MS Teams, Slack, Discord, Google Groups, etc. Please refrain from using IMs/Text Messages for the purposes of this project.

> **TIP**
>
> You can track communication for project development tasks as comments on the user story issues you create on the GitHub project board. That will help you keep the discussions topic specific and organized.

11. Meeting Plan

- **Team Meeting**: Identify the day(s) and time(s), mode, and location your team has agreed upon for regular meetings. “Modality” refers to how you are meeting (face-to-face), group video chat (like Zoom, Google Meet), etc.
- **Weekly meeting with TA**: You will also mention the meeting day, time and location (in-person or online) for the weekly meeting with your TA here.

- Please refer to **these instructions** for expectations in your weekly meeting with your TAs. Do read the entire section carefully.

12. Use Case Diagram

Create a high-level Use Case Diagram for your application. Your diagram must show at least 6 key features of your application.

This activity should help you identify the end users of the application and the ways they would interact with the system. You can reference the examples covered in class. This will also help you scope out the features of your application. You can refer to **this page** for more information on how to create a Use Case Diagram.

13. Wireframes

You will create wireframes for each page of the application.

Wireframes are low-fidelity visual representations of the user interface of your application. They are used to plan the layout of the application and to communicate the flow of the application to the team. You can use any tool of your choice to create the wireframes. Hand sketched wireframes are also acceptable. You can refer to **this** for more information on why and how to create wireframes.

Part B

There is no part B for this lab 🎉

Extra Credit

Identify at least five potential risks related to your project. For each risk, describe its severity and outline suitable mitigation strategies to address it. Make sure to add a note in your proposal that this is the submission for extra credit.

Submission Guidelines

Commit and upload your changes

1. Upload your file(s), containing all the requested information above, in the `milestone submissions` folder of your project repository.

2. Run the following commands in the root of the project git directory (that you created).

```
git add .
git commit -m "Adding project proposal"
git push
```

3. Go to Github and verify that all the files/folders for the lab have been successfully uploaded to the remote repository.

4. Submit a link to your GitHub repository to Canvas. Only 1 person per team needs to do this.

Regrade Requests

Please use this **link** to raise a regrade request if you think you didn't receive a correct grade.

Grading Specifications

Criteria	Not Attempted	Below Expectations	Meets Expectations	Exceeds Expectations
Overall Attempt	• Lab not attempted **OR** • Student did not attend the lab without prior approval	—	—	—

Criteria	Not Attempted	Below Expectations	Meets Expectations	Exceeds Expectations
Attendance & TA Review	—	• Did not attend full duration of lab **OR** • Work not reviewed by TA **OR** • Attendance not marked on Canvas	• Attended full duration of lab **AND** • Demonstrated work to TA **AND** • Attendance marked on Canvas	—
Submission Presence	—	• Work not uploaded to GitHub repository **OR** • Required files/folders missing **OR** • Files do not represent a genuine attempt **OR** • GitHub link to project proposal not submitted on Canvas	• All required files and folders created and uploaded to the repository **AND** • GitHub link to the project proposal is submitted on Canvas	—
Project Specification Responses (#1–7, #9–11)	—	• One or more required responses missing **OR** • Responses incomplete or unclear	• All requested details for items **#1–7 and #9–11** clearly included as specified in the writeup	—

Criteria	Not Attempted	Below Expectations	Meets Expectations	Exceeds Expectations
Repository Structure (#8)	—	• Required files/folders for item #8 missing or incorrectly structured	• All required files and folders created in the repository according to the instructions	—
Use Case Diagram (#12)	—	• Use case diagram missing **OR** • Does not represent required application features	• Use case diagram included representing at least 6 key features of the application	—
Wireframes (#13)	—	• Wireframes missing **OR** • Do not cover all application pages	• Clearly drawn wireframes included for all application pages (hand-drawn acceptable)	—
Extra Credit – Risk Analysis	—	—	—	• Meets all Meets Expectations criteria **AND** • Identifies 5 project risks • Lists severity level for each risk • Provides

Criteria	Not Attempted	Below Expectations	Meets Expectations	Exceeds Expectations
				appropriate mitigation strategies for each risk